

E-File System – Modification Requirement Document (Version 1.0)

Project: E-File Management System

Client: AVFU

Date: 24 July 2026

1. Purpose

This document outlines the additional functional requirements requested by the client for the E-File Management System. These modifications will improve security, file workflow, tracking, confidentiality, and approval processes.

2. Functional Modifications

2.1 Two-Factor Authentication (OTP)

Requirement

All users must complete **OTP (One-Time Password) verification** after entering valid login credentials.

Functional Flow

1. User enters Username and Password.
2. System validates credentials.
3. System generates an OTP.
4. OTP is sent to the registered mobile number and.
5. User enters OTP.
6. After successful verification, the user is allowed to access the system.

Notes

- OTP should expire after a configurable time (recommended: 5 minutes).
- OTP can be used only once.
- Multiple failed attempts should invalidate the OTP.

2.2 Removal of Approval Button During File Movement

Requirement

The **Approval** button will not be available during normal file movement.

Users will only have the option to:

- Forward File
- Add Remarks/Comments

The approval process will occur only at the final stage.

2.3 Final Approval / Release / Close

Requirement

Only the **creator (originator)** of the file shall have permission to:

- Approve
- Release
- Close

This option will become available **only after the file has completed the workflow and returned to the creator.**

Conditions

- File must reach the creator again.
- File must be in the creator's inbox.
- Only then will the buttons appear:
 - Approve
 - Release
 - Close

2.4 Circular File Workflow

Requirement

The file workflow shall follow a circular routing process.

Example Workflow

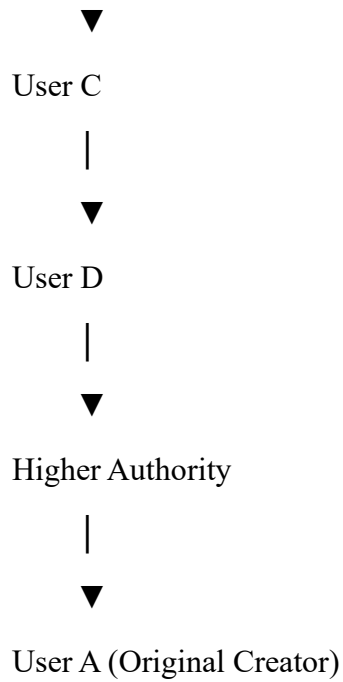
User A (Creator)

|



User B

|



After the final authority completes remarks/instructions, the file must be forwarded back to the original creator.

Once the file reaches the creator:

- Creator reviews the complete file.
- Creator can Approve.
- Creator can Release.
- Creator can Close.

2.5 Comment Visibility Rules

The system shall implement strict comment visibility.

During File Movement

Each user can view:

- Their own comments.
- Comments of the user from whom they directly received the file.

Users cannot view comments of users who receive the file later.

Example

$A \rightarrow B \rightarrow C \rightarrow D$

User C can view

- Comment of A
- Comment of B
- Own Comment

User C cannot view

- Comment of D
- Future comments
- Comments added after forwarding

Additional Rule

After User C forwards the file to User D:

User C can only view:

- Own comments

Previous users' comments become hidden.

2.6 Creator's Complete Access

Once the file returns to the creator, the creator shall have unrestricted access to all file information.

The creator can view:

- Complete forwarding history
- Complete activity log
- All comments
- All attachments
- Timeline of actions
- Current status
- Previous status
- Every movement of the file

No restrictions shall apply to the creator.

2.7 File Tracking History Module

A separate **File Tracking History** module shall be provided.

Filters

Users can search using:

- Today
- This Week
- This Month
- Last 3 Months
- Last 6 Months
- Custom Date Range

Tracking Information

Each record should display:

- File Number
- Subject
- Current Status
- Current Holder
- From User
- To User
- Forward Date & Time
- Action Date & Time
- Movement Timeline

Comment Visibility

Users can view:

- Their own comments only

Other users' comments must remain hidden.

However, users can still see:

- Who forwarded

- To whom
- Date & Time
- Status
- File movement history

2.8 Reopening Released/Closed Files

Released or Closed files may be reopened by an authorized user.

After Reopening

The same security rules shall apply.

Creator

Can view:

- Complete history
- Complete comments
- All attachments
- Entire workflow

Other Users

Can view only:

- Their own comments
- Their own actions
- Their own attachments (if applicable)

Other users' comments shall remain hidden.

2.9 File Priority / Type

The system shall support the following file categories.

File Type	Description
Confidential	One-to-one file movement only between designated sender and receiver. No additional forwarding unless explicitly permitted.
Urgent	High-priority file requiring immediate attention. Email notification must be sent.

Normal	Standard workflow without additional notification.
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2.10 Email Notification for Urgent Files

Whenever a file is marked as **Urgent**, the system shall automatically send an email notification.

Email notification should include:

- File Number
- Subject
- Sender Name
- Receiver Name
- Priority (Urgent)
- Date & Time
- Direct login link

Email should be triggered whenever:

- An urgent file is created.
- An urgent file is forwarded to another user.

2.11 Backtrack (Return File) Functionality

Requirement

The system shall provide a **Backtrack (Return File)** option at every stage of the file workflow. This feature allows the current holder of the file to return it to **any previously involved user** in the workflow for correction, clarification, or additional inputs.

Functional Rules

1. Every user receiving a file shall have the following actions:
 - **Forward**
 - **Backtrack (Return File)**
2. During backtracking, the system shall display a list of **all users who have previously handled the file**. The current user may select any one of these users as the backtrack recipient.
3. A **mandatory remark/comment** shall be entered before the file is backtracked, explaining the reason for returning the file.
4. The system shall record every backtrack action in the audit trail, including:

- Backtracked By
 - Backtracked To
 - Date & Time
 - Remarks/Reason
 - Action Type: **Backtrack**
5. Once the selected previous user receives the file, they may:
- Review and modify the file.
 - Add additional remarks or attachments.
 - Forward the file again through the normal workflow.
 - Backtrack the file again to any previously involved user, if required.
6. After the corrected file is forwarded, the workflow shall continue normally from that stage until it reaches the final authority and is eventually returned to the original creator for final approval/release/closure.

Additional Recommendations

To improve security, auditability, and usability, the following enhancements are recommended:

- Maintain a complete audit log for every file action (create, forward, reopen, release, close).
- Assign a unique transaction ID to each forwarding event for traceability.
- Display visual priority indicators (e.g., color-coded badges for Urgent and Confidential files).
- Prevent deletion of comments or forwarding history to preserve audit integrity.
- Record timestamps, user ID, IP address (optional), and action type for all critical operations.